SUN'S DIVISIBILITY CONJECTURE FOR $\sum_{k<n}(20k+17)\,W_k(1,-4)$: SQUAREFREE MODULI, THE SECOND LAYER, AND A REDUCTION VIA NEGATIVE-INDEX CONTINUATION

QEEDDD VERIFIABLE-MATHEMATICAL-REASONING-CORPUS PROJECT

Abstract. Let $a(n)=\sum_k(-4)^k\binom{n}{k}^2\binom{n-k}{k}^2=W_n(1,-4)$ and $S(n)=\sum_{k=0}^{n-1}(20k+17)a(k)$. In 2010 Z.-W. Sun conjectured that $n\mid S(n)$ for every $n\ge 1$ (OEIS A179524); the conjecture is listed as open in Sun's 2019 collection of open conjectures and in the `formal-conjectures` database (2026). We prove that $\mathrm{rad}(n)\mid S(n)$ for every $n\ge 1$; in particular *the conjecture holds for every squarefree n*. The proof combines a Lucas congruence, coming from a constant-term representation of $a(n)$, with a *reflection bridge* which converts $S(p)$ mod p into a truncation of Ramanujan's series $\sum_k(20k+3)\binom{2k}{k}^2\binom{4k}{2k}(-2^{10})^{-k}$, whose divisibility by $2h+1$ was proved by Sun using a WZ pair of Guillera and Zudilin. A second-order Frobenius expansion gives the *second-layer reduction* $S(p^2m)\equiv S(p^2)\sum_{t<m}a(t)\pmod{p^2}$; together with a p-adic computation of $S(p^2)$ mod p^2 for all primes $p\le 6000$ this proves the conjecture for every cube-free n whose square part has only prime factors ≤ 6000. We then introduce a *negative-index continuation*: in a ring of Laurent series one has $\sum_{n\ge1}(20n-17)\Lambda^{-n}=(3M+20)/M^2$ with $M=\Lambda-1$, which gives $S(N)\equiv-\mathrm{CT}(\Lambda^N\Psi)\pmod N$ for an explicit power series $\Psi\in\mathbb{Z}[[u,v]]$. Hence the full conjecture follows from a new two-index statement (Conjecture B) asserting that $\gcd(\alpha,\beta)$ divides the coefficient of $u^\alpha v^\beta$ in Ψ; its diagonal is Sun's conjecture. Using integral changes of variables we show that Conjecture B is equivalent to two explicit one-variable congruences. These statements, a Dwork-type congruence that would imply the conjecture, and the conjecture itself for all $n\le 10^6$ are verified by exact computation; a targeted search at moduli with large square part (motivated by the square-modulus counterexample to an Erdős conjecture in the same corpus) found no counterexample. **We do not prove the conjecture for all n**: it remains open when $p^3\mid n$, or when $p^2\,\|\,n$ with $p>6000$.

1. Introduction

For integers b,c put $W_n(b,c)=\sum_k\binom{n}{2k}^2\binom{2k}{k}^2b^{n-2k}c^k$. Since $\binom{n}{k}\binom{n-k}{k}=\binom{n}{2k}\binom{2k}{k}$, the sequence of OEIS A179524 is

$$a(n)=\sum_{k=0}^{\lfloor n/2\rfloor}(-4)^k\binom{n}{k}^2\binom{n-k}{k}^2=\sum_k(-4)^k\binom{n}{2k}^2\binom{2k}{k}^2=W_n(1,-4), \tag{1}$$

so $(a(n))_{n\ge0}=1,1,-15,-143,\dots$. Throughout we write

$$S(n)=\sum_{k=0}^{n-1}(20k+17)\,a(k)\qquad(n\ge1).$$

Conjecture 1.1 (Z.-W. Sun, July 1, 2010)**.** *For every positive integer n we have $n\mid S(n)$.*

The conjecture appears as "conjecture 4" in the OEIS entry A179524 [12], in the 2019 edition of Sun's *Open conjectures on congruences* [15] (as $\sum_{k=0}^{n-1}(20k+17)W_k(1,-4)\equiv0\pmod n$), and as a `research open` statement in the Lean database `formal-conjectures` (file `FormalConjectures/OEIS/179524.lean`, added on 2026-11-06) [3]. We found no published proof (see §11); the companion congruence $S(p)\equiv p\,(10(\frac{-1}{p})+7)\pmod{p^2}$ is also open.

Date: November 7, 2026.

The weights $20k+17$ are rigid: among all $(\alpha,\beta)\neq(0,0)$ with $0\le\alpha\le 200$ and $|\beta|\le 400$, the only pairs with $n \mid \sum_{k<n}(\alpha k+\beta)a(k)$ for all $n\le 49$ are the multiples $(20t,17t)$, $1\le t\le 10$ (exhaustive search). Note $17=20-3$, and $(20,3)$ is the pair in Ramanujan's series

$$\sum_{k\ge 0}\frac{(20k+3)\binom{2k}{k}^2\binom{4k}{2k}}{(-2^{10})^k}=\frac{8}{\pi}. \tag{2}$$

This "duality" $\alpha k+\beta=\alpha(k+1)-\beta_{\mathrm{R}}$ is the starting point of our method.

Main results. Let $\mathrm{rad}(n)$ denote the product of the distinct primes dividing n.

Theorem A (Theorem 4.1)**.** *For every prime p and every $m\ge 1$ we have $p\mid S(pm)$. Consequently $\mathrm{rad}(n)\mid S(n)$ for every $n\ge 1$, and Conjecture 1.1 holds for every squarefree n.*

The two ingredients are a Lucas congruence $a(pm+r)\equiv a(m)a(r) \pmod p$ (Lemma 2.2) and the prime case $p\mid S(p)$ (Proposition 3.4). For the latter we prove a reflection congruence (Lemma 3.2) which identifies $S(p) \bmod p$, up to a unit, with the truncation of (2) at $k=(p-1)/2$; the needed divisibility of that truncation is Sun's theorem [14, Thm. 1.2], proved with the WZ pair of Guillera [4] and Zudilin [16].

Theorem A ($'$, Theorem 5.2 and Corollary 5.3)**.** *For every prime p and $m\ge 1$, $S(p^2m)\equiv S(p^2)\sum_{t<m}a(t) \pmod{p^2}$, and $S(p^2)\equiv -(3w_{p^2-1}+20w_{p^2-2}) \pmod{p^2}$ for explicit integers w_m. We verified $p^2\mid S(p^2)$ for all primes $p\le 6000$. Consequently $n\mid S(n)$ for every cube-free n all of whose prime factors p with $p^2\mid n$ satisfy $p\le 6000$.*

To go beyond these moduli we introduce the series

$$\rho=\frac{uv}{1+u+v},\qquad G(w)=\sum_{l\ge 0}(-4)^l\binom{2l}{l}^2 w^{2l},\qquad \Psi=(3\rho+20\rho^2)\,G(uv)\in\mathbb{Z}[[u,v]],$$

and the two-index numbers $W_{a,b}=\sum_j(-4)^j\binom{2j}{j}^2\binom{a}{2j}\binom{b}{2j}$ (so $W_{k,k}=a(k)$).

Theorem B (Proposition 6.2, Theorems 6.4 and 6.7)**.** *(i) (Negative-index continuation) With $\Lambda=(1+u)(1+v)/(uv)$ one has $\sum_{n\ge 1}(20n-17)\Lambda^{-n}=3\rho+20\rho^2$ and, for all $N\ge 1$, $S(N)=-\operatorname{CT}(\Lambda^N\Psi)+20N\operatorname{CT}(\Lambda^N\rho G(uv))$.*

(ii) The following are equivalent: (a) $\gcd(\alpha,\beta)\mid[u^\alpha v^\beta]\Psi$ for all $(\alpha,\beta)\neq(0,0)$; (b) (Conjecture 6.5) $\gcd(\alpha,\beta)\mid\sum_{n\ge 1}(20n-17)W_{\alpha-n,\beta-n}$ for all $\alpha,\beta\ge 1$; (c) the one-variable congruences (I) *and* (II) *of Theorem 6.7.*

(iii) Each of (a)–(c) implies Conjecture 1.1; indeed the diagonal $\alpha=\beta=N$ of (b) is $N\mid S(N)$.

The congruences (I), (II) (one-variable sums of products of $\binom{2l}{l}^2$ and $\binom{2i}{i}$), Conjecture 6.5, and a Dwork-type congruence (Conjecture 7.1, which implies Conjecture 1.1 by Proposition 7.2) are supported by exact computations summarised in §9; in particular Conjecture 1.1 holds for all $n\le 10^6$.

What is not proved. Theorems A and A$'$ give $v_p(S(n))\ge\min(v_p(n),1)$ for all p, and $v_p(S(n))\ge\min(v_p(n),2)$ for $p\le 6000$, whereas Conjecture 1.1 requires $v_p(S(n))\ge v_p(n)$. The cases $p^3\mid n$, and $p^2\,\|\,n$ with $p>6000$, remain open. We isolate exactly what is missing in §5, §7 and §10.

2. Preliminaries

For a Laurent polynomial or series F we write $\operatorname{CT}F$ for its constant term and $[x^u]F$ for the coefficient of x^u. Binomial coefficients $\binom{X}{b}=X(X-1)\cdots(X-b+1)/b!$ are used for any X in a ring containing $\mathbb{Q}$, and $\binom{n}{b}=0$ for integers $0\le n<b$.

Lemma 2.1 (Constant-term representation)**.** *Let* $\Lambda_3 = (x+1-4z/x)\,(y+1+1/(yz))$*, an element of* $\mathbb{Z}[x^{\pm1}, y^{\pm1}, z^{\pm1}]$*. Then* $a(n) = \mathrm{CT}\,\Lambda_3^n$ *for all* $n \geq 0$*.*

Proof. For any w, $\mathrm{CT}_x(x+1+w/x)^n = \sum_j \frac{n!}{j!\,j!\,(n-2j)!} w^j = \sum_j \binom{n}{2j}\binom{2j}{j} w^j$. Apply this with $w = -4z$ and (in y) with $w = 1/z$, multiply, and take CT_z: only equal indices survive, giving $\sum_j (-4)^j \binom{n}{2j}^2 \binom{2j}{j}^2 = a(n)$. □

Lemma 2.2 (Lucas property)**.** *Let* p *be a prime,* $m \geq 0$ *and* $0 \leq r < p$*. Then* $a(pm+r) \equiv a(m)\,a(r) \pmod p$*.*

Proof. Since Λ_3 has integer coefficients, $\Lambda_3(x,y,z)^p \equiv \Lambda_3(x^p, y^p, z^p) \pmod p$, hence $\Lambda_3^{pm+r} \equiv \Lambda_3(x^p,y^p,z^p)^m\,\Lambda_3^r \pmod p$. Every monomial of Λ_3 has exponent vector in $\{-1,0,1\}^3$, so those of Λ_3^r lie in $[-r,r]^3 \subset (-p,p)^3$. Therefore $\mathrm{CT}\left(\Lambda_3(x^p,y^p,z^p)^m \Lambda_3^r\right) = \sum_v [x^{pv}]\Lambda_3(x^p,y^p,z^p)^m \cdot [x^{-pv}]\Lambda_3^r$ receives a contribution only from $v=0$, and equals $\mathrm{CT}(\Lambda_3^m)\,\mathrm{CT}(\Lambda_3^r)$. Conclude with Lemma 2.1. □

Lemma 2.3 (Newton expansion)**.** *Put* $w_m = \sum_j (-4)^j \binom{2j}{j}^2 \binom{m}{2j}\binom{2j}{m-2j}$*. Then* $a(k) = \sum_{m=0}^k \binom{k}{m} w_m$ *for* $k \geq 0$*, and for every* $n \geq 1$

$$S(n) = \sum_{m=0}^{n-1} w_m \left(20(m+1)\binom{n}{m+2} + (20m+17)\binom{n}{m+1}\right). \tag{3}$$

Proof. Counting ordered pairs of i-subsets of $\{1,\dots,k\}$ according to the size m of their union gives $\binom{k}{i}^2 = \sum_m \binom{m}{i}\binom{i}{m-i}\binom{k}{m}$; insert this with $i = 2j$ into (1). For (3) use $k\binom{k}{m} = (m+1)\binom{k}{m+1} + m\binom{k}{m}$ and $\sum_{k=0}^{n-1}\binom{k}{r} = \binom{n}{r+1}$. □

(Inverting the binomial transform in $\mathrm{CT}\,\Lambda_3^k = \mathrm{CT}(1+(\Lambda_3-1))^k$ shows $w_m = \mathrm{CT}(\Lambda_3-1)^m$; since $\Lambda_3 - 1$ also has integer coefficients and exponents in $\{-1,0,1\}^3$, the proof of Lemma 2.2 gives $w_{pm+r} \equiv w_m w_r \pmod p$ for $0 \leq r < p$. Lucas- and Dwork-type congruences for constant terms of powers of Laurent polynomials are studied systematically in [11, 1]. Divisibility results of the type $n \mid \sum_{k<n}(\alpha k+\beta)u_k$ for other Apéry-like sequences were proved, e.g., in [5, 14].)

3. The prime case via a reflection bridge

Lemma 3.1. *Let* p *be an odd prime and* $h = (p-1)/2$*.*

(a) For $0 \leq i \leq p-1$*:* $\binom{p-1}{i} \equiv (-1)^i$ *and* $\binom{p-2}{i} \equiv (-1)^i(i+1) \pmod p$*.*

(b) For $0 \leq a \leq p-1$ *and* $0 \leq b \leq p-1$*:* $\binom{p-1-a}{b} \equiv (-1)^b\binom{a+b}{b} \pmod p$*.*

(c) For $0 \leq j \leq h$*:* $\binom{2j}{j} \equiv (-4)^j\binom{h}{j} \pmod p$*.*

Proof. For $0 \leq b < p$ the polynomial $\binom{X}{b}$ has coefficients in $\mathbb{Z}_{(p)}$, so $X \equiv Y \pmod p$ in $\mathbb{Z}_{(p)}$ implies $\binom{X}{b} \equiv \binom{Y}{b}$. (a) and (b) follow from $p-1-a \equiv -1-a$ and $\binom{-1-a}{b} = (-1)^b\binom{a+b}{b}$ (with $a=0$, resp. $a=1$, for (a)). (c) follows from $\binom{2j}{j} = (-4)^j\binom{-1/2}{j}$ and $h \equiv -1/2 \pmod p$. □

Lemma 3.2 (Reflection bridge)**.** *Let* p *be an odd prime and* $h = (p-1)/2$*. Then, modulo* p*,*

$$w_{p-1} \equiv (-4)^{3h} \sum_{l=0}^{h} \frac{\binom{2l}{l}^2\binom{4l}{2l}}{(-1024)^l}, \qquad w_{p-2} \equiv (-4)^{3h} \sum_{l=0}^{h} \frac{l\binom{2l}{l}^2\binom{4l}{2l}}{(-1024)^l}.$$

Proof. For $m \in \{p-1, p-2\}$ the sum defining w_m may be taken over $0 \leq j \leq h$ (the other terms vanish). Substitute $j = h-l$ with $0 \leq l \leq h$, so that $2j = p-1-2l$. By Lemma 3.1(a), $\binom{p-1}{2j} \equiv 1$ and $\binom{p-2}{2j} \equiv 2j+1 \equiv -2l$. By (b) with $a = b = 2l$, $\binom{2j}{p-1-2j} = \binom{p-1-2l}{2l} \equiv \binom{4l}{2l}$; by (b) with $a = 2l$, $b = 2l-1$ ($l \geq 1$), $\binom{2j}{p-2-2j} = \binom{p-1-2l}{2l-1} \equiv -\binom{4l-1}{2l-1} = -\frac{1}{2}\binom{4l}{2l}$; for $l = 0$ the corresponding term of w_{p-2}

vanishes because $\binom{p-2}{p-1}=0$. Finally, (c) applied to j and to l gives $\binom{2j}{j}\equiv(-4)^j\binom{h}{l}\equiv(-4)^{j-l}\binom{2l}{l}=(-4)^{h-2l}\binom{2l}{l}$, whence $(-4)^j\binom{2j}{j}^2\equiv(-4)^{3h-5l}\binom{2l}{l}^2$. Multiplying out and using $(-4)^5=-1024$ gives both congruences (for w_{p-2} the factor is $(-2l)(-\frac{1}{2}\binom{4l}{2l})=l\binom{4l}{2l}$). □

Theorem 3.3 (Sun [14, Theorem 1.2, (1.5)]). *For every $n\ge 1$,*

$$4(2n+1)\binom{2n}{n}\;\Big|\;R(n):=\sum_{k=0}^{n}(20k+3)\binom{2k}{k}^2\binom{4k}{2k}(-2^{10})^{n-k}.$$

Sun's proof uses the WZ pair of Guillera and Zudilin [4, 16] for (2).

Proposition 3.4 (Prime case). *For every prime p we have $p\mid S(p)$.*

Proof. For $p=2$: $S(2)=17a(0)+37a(1)=54$. Let p be odd, $h=(p-1)/2\ge 1$. In (3) with $n=p$ we have $p\mid\binom{p}{r}$ for $1\le r\le p-1$, $\binom{p}{p}=1$ and $\binom{p}{r}=0$ for $r>p$, so only $m=p-2$ (first term) and $m=p-1$ (second term) survive modulo p:

$$S(p)\equiv 20(p-1)w_{p-2}+(20(p-1)+17)w_{p-1}\equiv-(20w_{p-2}+3w_{p-1})\pmod p.$$

By Lemma 3.2, $20w_{p-2}+3w_{p-1}\equiv(-4)^{3h}(-1024)^{-h}R(h)\pmod p$, and by Theorem 3.3 $p=2h+1$ divides $R(h)$. □

4. Proof of Theorem A

Theorem 4.1. *For every prime p and every integer $m\ge 1$, $p\mid S(pm)$. Hence $\operatorname{rad}(n)\mid S(n)$ for all $n\ge 1$; in particular $n\mid S(n)$ for every squarefree n.*

Proof. Writing $k=pq+r$ $(0\le q<m,\ 0\le r<p)$ and using Lemma 2.2 and $20(pq+r)+17\equiv 20r+17$,

$$S(pm)\equiv\sum_{q=0}^{m-1}a(q)\sum_{r=0}^{p-1}(20r+17)a(r)=S(p)\sum_{q=0}^{m-1}a(q)\equiv 0\pmod p$$

by Proposition 3.4. If $p\mid n$ then $n=pm$ and $p\mid S(n)$; thus every prime factor of n divides $S(n)$. □

5. The second layer

In this section $\Lambda=\Lambda_3$, p is a prime, $\Lambda^{(p)}=\Lambda(x^p,y^p,z^p)$ and $\Phi=(\Lambda^p-\Lambda^{(p)})/p$, a Laurent polynomial with integer coefficients and exponents in $[-p,p]^3$. For $v\in\mathbb{Z}^3$ and $s\ge 0$ put $\beta_v(s)=[x^v]\Lambda^s$ (so $\beta_0=a$), and write $v_m=3w_m+20w_{m-1}$ $(w_{-1}=0)$. The proof of Proposition 3.4 shows $S(p)\equiv -v_{p-1}\pmod p$, hence $p\mid v_{p-1}$.

Lemma 5.1 (Second-order Lucas). *(i) For $0\le r\le p-1$ and $q\ge 1$: $a(r+pq)\equiv a(r)a(q)+pq\,\operatorname{CT}\left(\Lambda^r\Phi\,(\Lambda^{(p)})^{q-1}\right)\pmod{p^2}$.*

(ii) For $v\in\{-1,0,1\}^3$, $0\le d\le p-2$ and $t\ge 0$: $\beta_v(d+pt)\equiv\beta_v(d)\,a(t)\pmod p$.

(iii) If the exponents of a Laurent polynomial F lie in $(-2p,2p)^3$, then for $s\ge 0$

$$\operatorname{CT}\left(F\,(\Lambda^{(p)})^s\right)=\sum_{v\in\{-1,0,1\}^3}[x^{-pv}]F\cdot\beta_v(s).$$

Proof. (i) $(\Lambda^{(p)}+p\Phi)^q\equiv(\Lambda^{(p)})^q+pq\,\Phi(\Lambda^{(p)})^{q-1}\pmod{p^2}$, and $\operatorname{CT}\left(\Lambda^r(\Lambda^{(p)})^q\right)=a(r)a(q)$ exactly, as in the proof of Lemma 2.2. (ii) $\Lambda^{d+pt}\equiv\Lambda^d(\Lambda^{(p)})^t\pmod p$ and $[x^v]\big(\Lambda^d(\Lambda^{(p)})^t\big)=\sum_w[x^{v-pw}]\Lambda^d\cdot\beta_w(t)$; the exponents of Λ^d lie in $[-d,d]^3$, while for $w\ne 0$ some coordinate of $v-pw$ has absolute value $\ge p-1>d$. (iii) $[x^{pv}](\Lambda^{(p)})^s=\beta_v(s)$, and $[x^{-pv}]F=0$ unless $v\in\{-1,0,1\}^3$. □

Theorem 5.2 (Second-layer reduction)**.** *For every prime p and every $m \geq 1$,*

$$S(p^2m) \equiv S(p^2)\sum_{t=0}^{m-1} a(t) \pmod{p^2}, \qquad S(p^2) \equiv -\big(3w_{p^2-1} + 20w_{p^2-2}\big) \pmod{p^2}.$$

Proof. Put $\sigma_m = \sum_{t<m} a(t)$, $A_p = \sum_{r<p} a(r)$, $B_p = \sum_{r<p} r\,a(r)$, $Q_1 = \sum_{r<p}(20r+17)\Lambda^r$ and, for $q \geq 1$, $E(q) = \mathrm{CT}\left(Q_1\Phi(\Lambda^{(p)})^{q-1}\right)$, $E(0) = 0$. Writing $k = r + pq$ with $0 \leq r < p$, $0 \leq q < pm$, Lemma 5.1(i) gives

$$\begin{aligned} S(p^2m) &\equiv \sum_{q<pm}\sum_{r<p}(20r+17+20pq)\big(a(r)a(q) + pq\ \mathrm{CT}(\Lambda^r\Phi(\Lambda^{(p)})^{q-1})\big) \\ &\equiv S(p)\sum_{q<pm} a(q) + 20pA_p\sum_{q<pm} q\,a(q) + p\sum_{q<pm} qE(q) \end{aligned}$$

modulo p^2. By Lemma 2.2, $\sum_{q<pm} a(q) \equiv A_p\sigma_m$ and $\sum_{q<pm} q\,a(q) \equiv B_p\sigma_m \pmod p$; as $p \mid S(p)$, the first term is $\equiv S(p)A_p\sigma_m \pmod{p^2}$. The exponents of $Q_1\Phi$ lie in $[-(2p-1), 2p-1]^3$, so by Lemma 5.1(iii) $E(q) = \sum_v g_v\beta_v(q-1)$ with $g_v = [x^{-pv}](Q_1\Phi)$. For $q = r'+pt$ with $1 \leq r' \leq p-1$ we have $q \equiv r'$ and, by Lemma 5.1(ii), $\beta_v(q-1) \equiv \beta_v(r'-1)a(t) \pmod p$; the terms with $r' = 0$ vanish modulo p. Hence $\sum_{q<pm} qE(q) \equiv \sigma_m\sum_v g_vC_v \pmod p$ with $C_v = \sum_{r'=1}^{p-1} r'\beta_v(r'-1)$. Altogether $S(p^2m) \equiv pK_p\sigma_m \pmod{p^2}$ with $K_p = S(p)A_p/p + 20A_pB_p + \sum_v g_vC_v$ independent of m; $m = 1$ gives $S(p^2) \equiv pK_p$.

For the second congruence use (3) with $n = p^2$. By Kummer's theorem $v_p\binom{p^2}{r} = 2 - v_p(r)$ for $1 \leq r < p^2$, so modulo p^2 only $r = jp$ $(1 \leq j \leq p)$ survive. For $1 \leq j \leq p-1$ the two terms with $r = jp$ give $\binom{p^2}{jp}\big(-v_{jp-1} + p\kappa_j\big)$ $(\kappa_j \in \mathbb{Z})$, and $v_{jp-1} \equiv w_{j-1}v_{p-1} \equiv 0 \pmod p$ by the Lucas property of w; these terms vanish modulo p^2. The terms with $r = p^2$ give $20(p^2-1)w_{p^2-2} + (20p^2-3)w_{p^2-1} \equiv -v_{p^2-1} \pmod{p^2}$. □

Corollary 5.3. *Let $n \geq 1$, and assume that every prime p with $p^2 \mid n$ satisfies $p^3 \nmid n$ and $p^2 \mid S(p^2)$. Then $n \mid S(n)$.*

Proof. If $p \,\|\, n$ use Theorem 4.1. If $p^2 \,\|\, n$ write $n = p^2m$; then $S(n) \equiv S(p^2)\sigma_m \equiv 0 \pmod{p^2}$ by Theorem 5.2. □

Proposition 5.4. *$p^2 \mid S(p^2)$ for every prime $p \leq 6000$. Consequently Conjecture 1.1 holds for every cube-free n such that every prime p with $p^2 \mid n$ is at most 6000.*

Proof. For $p \leq 443$ this is contained in the exact verification of $n \mid S(n)$ for $n \leq 200\,000$ (and for $p \leq 997$ in the run up to 10^6). For $449 \leq p \leq 6000$ we computed $3w_{p^2-1} + 20w_{p^2-2}$ modulo p^4, summing the terms $(-4)^j\, m!\,(2j)!^2/\big(j!^4(m-2j)!^2(4j-m)!\big)$ of w_m with exact p-adic valuations and unit parts modulo p^4 (program `ctest.c`); in every case its valuation is ≥ 2, so Theorem 5.2 gives $p^2 \mid S(p^2)$. The program was cross-checked against exact integer arithmetic for 17 prime powers $p^e \leq 1024$ (including $p = 2$). The second statement is Corollary 5.3. □

Remark 5.5 (Higher layers)**.** Numerically, $S(p^em) \equiv S(p^e)\sum_{t<m} a(t) \pmod{p^e}$ whenever $p^em \leq 6000$ and either $p \leq 13$, $e \leq 2$, or $p \leq 11$, $e = 3$, or $p \leq 5$, $e = 4$. Even the blockwise congruences $\sum_{r<p^e}(20r+17)a(r+p^et) \equiv S(p^e)a(t) \pmod{p^e}$ hold for $p \leq 13$, $e \leq 2$ and $p \leq 11$, $e = 3$ (with $p^e(t+1) \leq 6000$), and for $(p,e) \in \{(2,5),(2,6),(3,4),(3,5),(5,3)\}$ (with $p^e(t+1) \leq 2600$). For $e \leq 2$ the blockwise congruence also held for the weights 1, r, r^2 in place of $20r+17$; for $e \geq 3$ it is weight-sensitive: for $p \in \{3,7\}$, $e = 3$ it fails for weights $\alpha r + \beta$ with $p \nmid \sum_{r<p}(\alpha r+\beta)a(r)$ and held for all tested weights with p dividing this sum, whereas for $p = 3$, $e = 4$ some of the latter fail although Sun's weight $20r+17$ does not.

Conjecture 5.6 (Layer principle). *For every prime power p^e and every $m \geq 1$, $S(p^e m) \equiv S(p^e) \sum_{t<m} a(t) \pmod{p^e}$.*

Under Conjecture 5.6, Conjecture 1.1 is equivalent to $p^e \mid S(p^e)$ for all prime powers p^e: the "bottom layer". Theorem 5.2 is the case $e = 2$, and the bottom layer is proved for $e = 1$ (Proposition 3.4) and verified for $e = 2$, $p \leq 6000$.

6. Negative-index continuation and a two-index generalisation

Let $A = \mathbb{Z}[[u, v]]$, $B = \bigcup_{N \geq 0} (uv)^{-N} A$, and let $\mathrm{CT}\colon B \to \mathbb{Z}$ be the coefficient of $u^0 v^0$. The derivations $\theta_u = u\,\partial_u$, $\theta_v = v\,\partial_v$ act on B and $\mathrm{CT}(\theta_u H) = \mathrm{CT}(\theta_v H) = 0$ for $H \in B$. Put

$$\Lambda = \frac{(1+u)(1+v)}{uv}, \quad \rho = \frac{uv}{1+u+v}, \quad \gamma_l = (-4)^l \binom{2l}{l}^2,$$
$$G(w) = \sum_{l \geq 0} \gamma_l w^{2l}, \qquad \Psi = (3\rho + 20\rho^2) G(uv).$$

Lemma 6.1. *$a(k) = \mathrm{CT}\left(\Lambda^k G(uv)\right)$ for all $k \geq 0$.*

Proof. $\mathrm{CT}\left((1+u)^k (1+v)^k (uv)^{2l-k}\right) = \binom{k}{k-2l}^2 = \binom{k}{2l}^2$, and $a(k) = \sum_l \gamma_l \binom{k}{2l}^2$. □

Proposition 6.2 (Negative-index continuation). *(i) $\Lambda - 1$ is invertible in B with inverse ρ, and $\sum_{n \geq 1} (20n - 17) \Lambda^{-n} = 3\rho + 20\rho^2$ in A.*

(ii) For every $N \geq 1$, $S(N) = -\,\mathrm{CT}(\Lambda^N \Psi) + 20N\ \mathrm{CT}\left(\Lambda^N \rho\, G(uv)\right)$.

Proof. (i) $\Lambda - 1 = (1 + u + v)/(uv)$. With $t = \Lambda^{-1} = uv/((1+u)(1+v)) \in uvA$ the series converges, $t/(1-t) = \rho$, and $\sum_{n \geq 1} (20n - 17) t^n = (3t + 17t^2)/(1-t)^2 = 3\rho + 20\rho^2$. (ii) In $\mathbb{Z}[T]$ one checks $(T-1)^2 \sum_{k=0}^{N-1} (20k + 17) T^k = (3T + 17)(1 - T^N) + 20NT^N (T - 1)$. Substitute $T = \Lambda$ and multiply by $\rho^2 = (\Lambda - 1)^{-2}$; since $(3\Lambda + 17)\rho^2 = (3(\Lambda - 1) + 20)\rho^2 = 3\rho + 20\rho^2$ we get $\sum_{k<N} (20k + 17) \Lambda^k = (3\rho + 20\rho^2)(1 - \Lambda^N) + 20N\Lambda^N \rho$. Multiply by $G(uv)$, take CT, and use Lemma 6.1 and $\mathrm{CT}\,\Psi = 0$ ($\Psi \in uvA$). □

Thus $3\rho + 20\rho^2 = -\sum_{k \leq -1} (20k + 17) \Lambda^k$ is Sun's weight continued to negative indices; Proposition 6.2 expresses $S(N)$ through this "negative half".

Let $\mathcal{D}_A = \theta_u A + \theta_v A$ ("formal derivatives").

Lemma 6.3. *$F = \sum f_{\alpha\beta} u^\alpha v^\beta \in A$ lies in $\mathcal{D}_A$ iff $f_{00} = 0$ and $\gcd(\alpha, \beta) \mid f_{\alpha\beta}$ for all $(\alpha, \beta) \neq (0, 0)$.*

Proof. Necessity is clear. Conversely choose integers with $\alpha s_{\alpha\beta} + \beta t_{\alpha\beta} = f_{\alpha\beta}$ and put $J_1 = \sum s_{\alpha\beta} u^\alpha v^\beta$, $J_2 = \sum t_{\alpha\beta} u^\alpha v^\beta$; then $F = \theta_u J_1 + \theta_v J_2$. □

Theorem 6.4 (Reduction). *If $\Psi \in \mathcal{D}_A$, then $N \mid S(N)$ for every $N \geq 1$.*

Proof. Write $\Psi = \theta_u J_1 + \theta_v J_2$ with $J_i \in A$. Then

$$\mathrm{CT}(\Lambda^N \theta_u J_1) = -\,\mathrm{CT}(\theta_u(\Lambda^N) J_1) = -N\ \mathrm{CT}(\Lambda^{N-1} \theta_u(\Lambda) J_1) \in N\mathbb{Z},$$

similarly for v, and $20N\ \mathrm{CT}(\Lambda^N \rho G(uv)) \in N\mathbb{Z}$. Apply Proposition 6.2(ii). □

Conjecture 6.5 (Two-index generalisation). *Let $W_{a,b} = \sum_j \gamma_j \binom{a}{2j} \binom{b}{2j}$. For all $\alpha, \beta \geq 1$,*

$$\gcd(\alpha, \beta) \;\Big|\; \mathcal{B}(\alpha, \beta) := \sum_{n=1}^{\min(\alpha,\beta)} (20n - 17)\, W_{\alpha-n, \beta-n}.$$

Since $W_{k,k} = a(k)$, $\mathcal{B}(N, N) = \sum_{k=0}^{N-1} (20N - 20k - 17) a(k) \equiv -S(N) \pmod{N}$; so the diagonal of Conjecture 6.5 is Conjecture 1.1.

6.1. **Integral changes of variables.**

Lemma 6.6. *Let $P \in u\mathbb{Z}[[u]]$ with $P'(0)=1$ and $q \in 1+u\mathbb{Z}[[u]]$. The substitution $\iota\colon \mathbb{Z}[[s,t]] \to A$, $F(s,t) \mapsto F(P(u), v\,q(u))$, is a ring isomorphism, and with the unit $c = \theta_u(\log P) = uP'/P \in 1+u\mathbb{Z}[[u]]$ we have $\mathcal{D}_A = c\cdot\iota(\mathcal{D}_{\mathbb{Z}[[s,t]]})$. The same holds with the roles of u and v interchanged.*

Proof. ι is an isomorphism since its linear part is the identity. Put $e = \theta_u(\log q) \in u\mathbb{Z}[[u]]$. The chain rule gives, for $F \in \mathbb{Z}[[s,t]]$, $\theta_u(\iota F) = c\,\iota(\theta_s F) + e\,\iota(\theta_t F)$ and $\theta_v(\iota F) = \iota(\theta_t F)$. Since c, e depend on u only, $c\,\iota(\theta_s F) = \theta_u(\iota F) - \theta_v(e\,\iota F)$ and $c\,\iota(\theta_t F) = \theta_v(c\,\iota F)$, so $c\,\iota(\mathcal{D}_{\mathbb{Z}[[s,t]]}) \subset \mathcal{D}_A$. Conversely, writing $J_i = \iota F_i$, $\theta_u J_1 + \theta_v J_2 = c\,\iota\big(\theta_s F_1 + \theta_t(\tilde e F_1 + \tilde c F_2)\big)$ with $\tilde e = \iota^{-1}(e/c)$, $\tilde c = \iota^{-1}(1/c) \in \mathbb{Z}[[s]]$ (functions of s alone commute with θ_t). □

Theorem 6.7 (Equivalent forms)**.** *For $a \ge 1$, $b \ge 1$ and $0 \le l \le (a-1)/2$ put $i = a-1-2l$, $\sigma_l(a) = \gamma_l\binom{2i}{i}(10i+3)$,*

$$Y(a) = \sum_{l=0}^{\lfloor (a-1)/2\rfloor} \sigma_l(a) = \sum_l (-4)^l \binom{2l}{l}^2 \big(10(a-2l)-7\big)\binom{2a-4l-2}{a-2l-1}, \qquad T(a,b) = \sum_{l=0}^{\lfloor (b-1)/2\rfloor} \sigma_l(a).$$

The following are equivalent:

(a) $\Psi \in \mathcal{D}_A$, *i.e.* $\gcd(\alpha,\beta) \mid [u^\alpha v^\beta]\Psi$ *for all* $\alpha,\beta \ge 1$;
(b) Conjecture 6.5;
(c) (I) $a \mid Y(a)$ *for all* $a \ge 1$, *and* (II) $\gcd(a,b) \mid T(a,b)$ *for all* $1 \le b < a$.

Proof. (a)⇔(b). Let $\Psi'(U,V) = \sum_{\alpha,\beta} \mathcal{B}(\alpha,\beta)U^\alpha V^\beta = F(UV)\Gamma(U,V)$ with $F(W) = \sum_{n\ge1}(20n-17)W^n$ and $\Gamma = \sum_{a,b} W_{a,b}U^aV^b = \sum_j \gamma_j (UV)^{2j}((1-U)(1-V))^{-2j-1}$. Apply Lemma 6.6 twice (to $U = u/(1+u)$, then to $V = v/(1+v)$); the Jacobian factor is $1/((1+u)(1+v))$. Under this substitution $UV/((1-U)(1-V)) = uv$, $1/(1-U) = 1+u$, and $UV/(1-UV) = \rho$, so $F(UV) = 3\rho + 20\rho^2$ and $\Gamma = (1+u)(1+v)G(uv)$. Hence $\iota(\Psi')\cdot\frac{1}{(1+u)(1+v)} = \Psi$ and Lemma 6.6 gives $\Psi' \in \mathcal{D} \iff \Psi \in \mathcal{D}_A$; by Lemma 6.3 the former is (b).

(a)⇔(c). Apply Lemma 6.6 with $P(u) = u(1+u)$, $q = (1+u)^{-1}$, i.e. $\varphi = u(1+u)$, $\chi = v/(1+u)$, $c = (1+2u)/(1+u)$. Let $E(\varphi) = (1+4\varphi)^{-1/2} = \sum_i (-1)^i\binom{2i}{i}\varphi^i$ and

$$\Theta(\varphi,\chi) = \varphi\chi\,\frac{10\chi + (3-7\chi)E(\varphi)}{(1+\chi)^2}\,G(\varphi\chi) \in \mathbb{Z}[[\varphi,\chi]].$$

Since $1+4\varphi = (1+2u)^2$ we have $\iota(E) = 1/(1+2u)$ and $\iota(\varphi\chi) = uv$; a direct computation (using $1+\chi = (1+u+v)/(1+u)$ and $10\chi + \frac{3-7\chi}{1+2u} = \frac{3(1+u)+v(3+20u)}{(1+u)(1+2u)}$) gives $\iota(\Theta)\,c = \frac{uv\,(3(1+u+v)+20uv)}{(1+u+v)^2}G(uv) = \Psi$. So (a) is equivalent to $\Theta \in \mathcal{D}$. Now $[\varphi^i\chi^j]\frac{10\chi+(3-7\chi)E}{(1+\chi)^2} =: \kappa_{ij}$ equals $3(-1)^j$ for $i=0$ and $(-1)^{i+j}\binom{2i}{i}(10j+3)$ for $i \ge 1$; hence $[\varphi^a\chi^b]\Theta = \sum_l \gamma_l \kappa_{a-1-2l,\,b-1-2l}$ (indices ≥ 0). Put $i = a-1-2l$, $j = b-1-2l$, so $j-i = b-a$ and $(-1)^{i+j} = (-1)^{a+b}$. In all cases $\kappa_{ij} = (-1)^{a+b}\binom{2i}{i}(10i+3) + 10(j-i)(-1)^{a+b}\binom{2i}{i}[i \ge 1]$. If $b \ge a$ the summation range is $2l+1 \le a$, so $[\varphi^a\chi^b]\Theta \equiv (-1)^{a+b}Y(a) \pmod{b-a}$; if $b < a$ it is $2l+1 \le b$, so $[\varphi^a\chi^b]\Theta \equiv (-1)^{a+b}T(a,b) \pmod{a-b}$. As $\gcd(a,b)$ divides $a-b$, Lemma 6.3 shows that $\Theta \in \mathcal{D}$ iff $\gcd(a,b) \mid Y(a)$ for $b \ge a$ (which for $b = a$ is (I) and for $b > a$ follows from (I)) and $\gcd(a,b) \mid T(a,b)$ for $b < a$ (which is (II)). □

Remark 6.8. (I) is the statement $\int_0^x (3+8t)(1-4t)^{-3/2}\,G(t)\,dt \in \mathbb{Z}[[x]]$, since $\sum_a Y(a)x^{a-1} = (3+8x)(1-4x)^{-3/2}G(x)$. A search for a closed form $A(x)G(x) + B(x)G'(x)$ of this integral with A, B in a natural class of algebraic functions, and for a linear relation with polynomial coefficients among the integral, G and G' (degree ≤ 30), found none; the integrality appears to be of genuinely arithmetic nature.

Remark 6.9 (Three variables)**.** With Λ_3 as in Lemma 2.1, $X = 1/x$, $Y = 1/y$ and $M = \Lambda_3 - 1$, the same computation in $\mathbb{Z}[z^{\pm 1}][[X, Y]]$ gives $R_0 := (3M + 20)/M^2 = \sum_{n\ge 1}(20n - 17)\Lambda_3^{-n}$ and $S(N) = -\operatorname{CT}(\Lambda_3^N R_0) + 20N \operatorname{CT}(\Lambda_3^N/M)$; so $N \mid S(N)$ would also follow if all coefficients of $X^a Y^b z^c$ in R_0 were divisible by $\gcd(a, b, c)$, which we checked for $a + b \le 12$. A closely related three-index strengthening of Conjecture 6.5 is

$$\gcd(\alpha, \beta, c) \;\Big|\; \sum_{n\ge 1}(20n - 17)\,[z^c]\,T_{\alpha-n}(1, -4z)\,T_{\beta-n}(1, 1/z) \qquad (\alpha, \beta \ge 1,\ c \in \mathbb{Z}),$$

where $T_m(b, c) = \sum_j \binom{m}{2j}\binom{2j}{j} b^{m-2j} c^j$ are the generalised central trinomial coefficients; its z^0-slice is exactly Conjecture 6.5 because $\operatorname{CT}_z\, T_a(1, -4z)T_b(1, 1/z) = W_{a,b}$. We verified it for $1 \le \alpha, \beta \le 100$, all c, and the p-parts of $\gcd(\alpha, \beta, c)$ for all primes $p \le 47$.

7. A Dwork-type congruence

Conjecture 7.1. *For every prime p there is $\lambda_p \in p\mathbb{Z}_{(p)}$ such that for all $n \ge 1$*

$$S(pn) \equiv \lambda_p\, S(n) \pmod{p^{v_p(n)+1}}.$$

Numerically one may take $\lambda_p = S(p)/17$ for $p \ne 17$ and $\lambda_{17} = S(34)/S(2)$ (§9); for $n = 1$ the congruence is consistent with Proposition 3.4.

Proposition 7.2. *Conjecture 7.1 implies Conjecture 1.1.*

Proof. Fix p; we show $v_p(S(n)) \ge v_p(n)$ by induction on $v_p(n)$, the case $v_p(n) = 0$ being trivial. If $n = pn'$ then $v_p(S(n)) \ge \min\big(v_p(n') + 1,\ v_p(\lambda_p) + v_p(S(n'))\big) \ge v_p(n') + 1$. □

By the Lucas property, Conjecture 7.1 holds modulo p (with both sides $\equiv 0$); the content is the higher-order term. Heuristically, in the language of Beukers–Vlasenko [1, 2] it says that a specific element of a Dwork crystal has vanishing unit-root projection for every p, which one would expect to come from the CM/Ramanujan structure behind (2); we have not made this precise.

8. Square-heavy moduli and an Erdős analogy

In companion manuscripts of the same corpus [8, 9, 10], Erdős's conjecture that $\sum_{n\ge 1} f(n)/n \ne 0$ for every convergent "Erdős function" f modulo q is shown to be *false* for $q = 105^2$ and *true* for every non-square q; the failures are confined to square moduli by a 2-adic transfer principle across the divisor layers $\gcd(n, q) = q/d$. Since Theorem A covers exactly the squarefree moduli, it is natural to ask whether Sun's conjecture could fail at moduli with a large square part. The two problems are of different nature (vanishing of a linear form in logarithms, decided by parity, versus p-adic valuations of a binomial sum), and we found no way to transfer the 2-adic argument itself. The layer viewpoint, however, led to Theorem 5.2, and we used the analogy to direct a counterexample search:

(a) exact verification of $n \mid S(n)$ for all $n \le 10^6$; this includes $105^2, 165^2, 195^2, 315^2, 495^2, 735^2$, all prime squares p^2 with $p < 1000$ and all prime powers up to that bound;
(b) for 1175 further square-heavy $n \le 10^7$ (all prime powers $p^e > 2 \cdot 10^5$ with $e \ge 2$, all products p^2q^2 with $p < 200 < q < 2000$ in range, and squares of odd squarefree numbers such as $1155^2, 1365^2, \dots$), the test $v_p(3w_{n-1} + 20w_{n-2}) \ge v_p(n)$ for every p with $p^2 \mid n$. Since $c_{n-1} = \sum_{k=1}^{n}(-1)^{n-k}\binom{n-1}{k-1}S(k)/k$, a failure would exhibit a counterexample $k \le n$.

No counterexample occurred.

9. Computations

All computations use exact integer arithmetic; the script is reproduced in Appendix A and its complete output is in the file `verify_log.txt`.

(1) $n \mid S(n)$ for all $1 \le n \le 10^6$ (the script in Appendix A reproduces the range $n \le 200\,000$ in about two minutes; the larger range uses the same code). Here $a(n)$ is computed by the order-four recurrence $\sum_{i=0}^{4} p_i(n)a(n+i) = 0$ listed in the script (found by linear algebra and cross-checked against (1) for $n < 400$; every division is exact).
(2) Lemmas 2.1, 2.2 ($p < 40$, $n < 600$), 2.3, the bridge congruences of Lemma 3.2 for all primes $3 \le p < 400$, and Theorem 3.3 for $1 \le n \le 150$.
(3) The identities of Proposition 6.2 and the coordinate identities in the proof of Theorem 6.7 (symbolically).
(4) Theorem 6.7(a) for $1 \le \alpha, \beta \le 110$; Conjecture 6.5 for $1 \le \alpha, \beta \le 120$; (I) for $a \le 700$ (and, separately, $a < 1500$); (II) for $1 \le b < a \le 160$. For comparison, in the three-variable check of Remark 6.9 replacing $(20, 17)$ by $(20, 16)$ or $(1, 0)$ produces 791, resp. 1674, failures among 129 600 coefficients, and in Theorem 6.7(a) replacing $(3, 20)$ by $(3, 19)$ fails already at $(\alpha, \beta) = (2, 2)$.
(5) Conjecture 7.1 for $3 \le p \le 31$ and $pn \le 1500$.
(6) Theorem 5.2 and Conjecture 5.6 as described in Remark 5.5; Proposition 5.4 for $p \le 6000$; the square-heavy search of §8.

10. Outlook

(a) *Prime powers.* By Proposition 7.2 and Theorem 6.7, it suffices to prove Conjecture 7.1, or the one-variable congruences (I) and (II); alternatively the layer principle (Conjecture 5.6) together with the bottom layers $p^e \mid S(p^e)$. For $e = 2$, numerically the analogue of Lemma 3.2 with p replaced by $N = p^2$ (both congruences, modulo p^2) holds for $p \in \{7, 11, 13, 17, 19, 31, 37\}$ but fails for $p \in \{3, 5, 23, 29\}$; where it holds, Theorem 3.3 would give $p^2 \mid S(p^2)$, but we have no proof of it. Both reduce, at a single prime, to the prime case proved here; what is missing is control of the second-order terms of the Lucas congruences. Two natural tools are the Dwork crystals of Beukers–Vlasenko [1, 2] and q-analogues with "creative microscoping" [6], which handle all cyclotomic factors $\Phi_d(q)$, $d \mid n$, simultaneously.
(b) *OEIS A179537.* Sun's companion conjecture $n \mid \sum_{k<n}(42k + 37)(-1)^k W_k(1, -16)$ has the same shape, with Ramanujan partner $\sum(42k + 5)\binom{2k}{k}^3/4096^k = 16/\pi$, whose p-adic analogue modulo p^4 was proved by Hu and Mao [7]; its Lucas property is proved exactly as Lemma 2.2. The corresponding prime-case bridge has not been carried out here, so we make no claim about it.

11. Status of the result

We record, as precisely as we can, what has and has not been achieved.

- *Named prior problem:* Conjecture 1.1 is Sun's conjecture of 2010 (OEIS A179524), stated in the original form.
- *Open status:* it is listed as open in [15] (2019) and in [3] (November 2026); the OEIS-Open paper [13] (August 2026), which reports language-model resolutions of 147 OEIS conjectures, does not mention A179524 or A179537. Literature searches (arXiv, journal databases reachable to us) found no proof. Absence of evidence is not proof of priority.
- *Completeness:* Theorem A is an unconditional theorem for *all squarefree* n (more precisely $\mathrm{rad}(n) \mid S(n)$ for all n), and Proposition 5.4 extends it to cube-free n whose square part

involves only primes ≤ 6000. **The full conjecture is not proved**: the remaining cases are reduced to Conjectures 7.1, 6.5 and 5.6, which remain open.

- *Non-triviality:* the prime case is not a re-expression of a known theorem: it requires the reflection congruence of Lemma 3.2, which maps the Newton coefficients w_{p-1}, w_{p-2} of a onto the Ramanujan truncation, combined with Sun's WZ-based divisibility.

Appendix A. Verification script

The following script (also distributed as `verify.py`) performs all checks of §9; run `python3 verify.py 200000`.

```
#!/usr/bin/env python3
"""Verification script for the paper
"Sun's divisibility conjecture for sum_{k<n}(20k+17) W_k(1,-4): the squarefree case
 and a reduction via negative-index continuation".
All checks use exact integer arithmetic.  Usage:  python3 verify.py [NMAX]
"""
import sys, time
from math import comb, gcd
import sympy as sp

NMAX = int(sys.argv[1]) if len(sys.argv) > 1 else 50000
ok_all = True
def report(name, ok):
    global ok_all
    ok_all &= ok
    print(("PASS " if ok else "FAIL ") + name, flush=True)

def a_direct(n):  # a(n) = sum_k (-4)^k C(n,k)^2 C(n-k,k)^2
    return sum((-4)**k * comb(n, k)**2 * comb(n-k, k)**2 for k in range(n//2 + 1))
def w(m):          # Newton coefficients
    return sum((-4)**j * comb(2*j, j)**2 * comb(m, 2*j) * comb(2*j, m-2*j)
               for j in range(m//2 + 1) if 0 <= m-2*j <= 2*j)

# 1. constant-term representation a(n) = CT(Lambda^n)
x, y, z = sp.symbols('x y z')
Lam = (x + 1 - 4*z/x) * (y + 1 + 1/(y*z))
def CT(expr):
    p = sp.Poly(sp.expand(expr * (x*y*z)**40), x, y, z)
    return p.coeff_monomial(x**40 * y**40 * z**40)
report("Lemma 2.1: a(n)=CT(Lambda^n), n<=8", all(CT(Lam**n) == a_direct(n) for n in range(9)))

# 2. Newton expansion and the formula for S(n)
okN = all(a_direct(k) == sum(w(m)*comb(k, m) for m in range(k+1)) for k in range(60))
S = lambda n: sum((20*k+17)*a_direct(k) for k in range(n))
okS = all(S(n) == sum(w(m)*(20*(m+1)*comb(n, m+2) + (20*m+17)*comb(n, m+1)) for m in range(n))
          for n in range(1, 60))
report("Lemma 2.3: Newton expansion and S(n) formula (k,n<60)", okN and okS)

# 3. Lucas property a(pm+r) = a(m)a(r) mod p
A = [a_direct(n) for n in range(600)]
okL = all((A[p*m+r] - A[m]*A[r]) % p == 0
          for p in sp.primerange(2, 40) for m in range(600//p) for r in range(p) if p*m+r < 600)
report("Lemma 2.2: Lucas property (p<40, n<600)", okL)

# 4. bridge congruences (Lemma 3.2) and Sun's theorem (EJC 2013, Thm 1.2) and S(p)=0 mod p
okB = True
for p in sp.primerange(3, 400):
    h = (p-1)//2; inv = pow(-4, -1, p); c0 = pow(-4, 3*h, p)
    B1 = sum(c0*pow(inv, 5*l, p)*comb(2*l, l)**2*comb(4*l, 2*l) for l in range(h+1)) % p
    B2 = sum(c0*pow(inv, 5*l, p)*l*comb(2*l, l)**2*comb(4*l, 2*l) for l in range(h+1)) % p
    okB &= (w(p-1) - B1) % p == 0 and (w(p-2) - B2) % p == 0
report("Lemma 3.2: bridge congruences for w_{p-1}, w_{p-2} (3<=p<400)", okB)
```

```
R = lambda h: sum((20*l+3)*comb(2*l, l)**2*comb(4*l, 2*l)*(-1024)**(h-l) for l in range(h+1))
report("Theorem 3.3 (Sun): 4(2h+1)C(2h,h) | R(h) (1<=h<=150)",
       all(R(h) % (4*(2*h+1)*comb(2*h, h)) == 0 for h in range(1, 151)))

# 5. partial-sum identity and negative-index continuation (symbolic)
T, s, u, v = sp.symbols('T s u v')
R0 = (3*T + 17)/(T - 1)**2
okP = all(sp.simplify(sum((20*k+17)*T**k for k in range(N)) - (R0 - T**N*(R0 - 20*N/(T-1)))) == 0
          for N in range(1, 9))
ser = sp.expand(sp.series(R0.subs(T, 1/s), s, 0, 15).removeO())
okNeg = all(ser.coeff(s, n) == 20*n - 17 for n in range(1, 15))
report("Prop 6.2: partial-sum identity (N<=8) and sum_{n>=1}(20n-17)T^{-n}=R0(T)", okP and okNeg)

# 6. coordinate identity of Theorem 5.4: Theta(phi(u),chi(u,v)) * (1+2u)/(1+u) = 3 rho + 20 rho^2
phi = u*(1+u); chi = v/(1+u); E = 1/(1+2*u)
Theta = phi*chi*(10*chi + (3 - 7*chi)*E)/(1 + chi)**2
rho = u*v/(1 + u + v)
report("Thm 6.7 proof: coordinate identity for (phi,chi)", sp.simplify(Theta*(1+2*u)/(1+u) - (3*rho + 20*rho
     **2)) == 0
       and sp.expand(1 + 4*phi - (1+2*u)**2) == 0)

# 7. Theorem 6.7(a) in (u,v)-coordinates: gcd(al,be) | [u^al v^be] Psi
gam = [(-4)**l*comb(2*l, l)**2 for l in range(400)]
def rho_coef(k, al, be):   # [u^al v^be] rho^k
    if al < k or be < k: return 0
    m = al + be - 2*k
    return (-1)**m * comb(k + m - 1, m) * comb(m, al - k)
def g(al, be): return 3*rho_coef(1, al, be) + 20*rho_coef(2, al, be)
def Bcoef(al, be): return sum(gam[l]*g(al - 2*l, be - 2*l) for l in range(min(al, be)//2 + 1))
M = 110
report("Thm 6.7(a): gcd(al,be) | [u^al v^be]Psi for 1<=al,be<=%d" % M,
       all(Bcoef(al, be) % gcd(al, be) == 0 for al in range(1, M+1) for be in range(1, M+1)))


# 7b. two-index form (Conjecture 6.5): gcd(al,be) | sum_{n>=1}(20n-17) W_{al-n,be-n}
def W2(a, b): return sum(gam[j]*comb(a, 2*j)*comb(b, 2*j) for j in range(min(a, b)//2 + 1))
W2tab = [[W2(a, b) for b in range(121)] for a in range(121)]
def B2(al, be): return sum((20*n - 17)*W2tab[al-n][be-n] for n in range(1, min(al, be) + 1))
report("Conjecture 6.5 (two-index form) for 1<=al,be<=120 and diagonal = -S(N) mod N",
       all(B2(al, be) % gcd(al, be) == 0 for al in range(1, 121) for be in range(1, 121))
       and all((B2(N, N) + S(N)) % N == 0 for N in range(1, 60)))
U_, V_ = sp.symbols('U V')
# coordinate change U=u/(1+u), V=v/(1+v): Psi'(U,V) dlogU dlogV = Psi(u,v) dlogu dlogv
rhoUV = (u/(1+u))*(v/(1+v))/(1 - (u/(1+u))*(v/(1+v)))
GamFac = (1+u)*(1+v)      # Gamma(U,V) = G(uv)/((1-U)(1-V)) with 1/(1-U)=1+u
report("Thm 6.7 proof: change (U,V)->(u,v) identity",
       sp.simplify(rhoUV - u*v/(1+u+v)) == 0 and
       sp.simplify((3*rhoUV + 20*rhoUV**2)*GamFac/((1+u)*(1+v)) - (3*rho + 20*rho**2)) == 0)

# 8. one-variable congruences (I) and (II)
def sig(a, l):
    i = a - 2*l - 1
    return gam[l]*comb(2*i, i)*(10*i + 3)
Y = lambda a: sum(sig(a, l) for l in range((a-1)//2 + 1))
Tt = lambda a, b: sum(sig(a, l) for l in range((b-1)//2 + 1))
report("Thm 6.7 (I): a | Y(a) for a<=700", all(Y(a) % a == 0 for a in range(1, 701)))
report("Thm 6.7 (II): gcd(a,b) | T(a,b) for 1<=b<a<=160",
       all(Tt(a, b) % gcd(a, b) == 0 for a in range(2, 161) for b in range(1, a)))

# 9. Sun's conjecture n | S(n) for n<=NMAX, using the order-4 recurrence of a(n)
def p0(n): return -4225*(n+1)**2*(n+2)**2*(240*n**3+2160*n**2+6483*n+6490)
def p1(n): return -2*(n+2)**2*(91680*n**5+1146000*n**4+5647266*n**3+13701286*n**2+16361708*n+7698385)
def p2(n): return -2*(16080*n**7+289440*n**6+2219001*n**5+9390796*n**4+23689644*n**3+35620787*n**2+29562269*n
     +10448048)
def p3(n): return 2*(n+3)*(480*n**6+7920*n**5+53646*n**4+190760*n**3+375516*n**2+388401*n+165272)
```

```
def p4(n): return -(n+3)*(n+4)**3*(240*n**3+1440*n**2+2883*n+1927)
t0 = time.time()
win = [a_direct(n) for n in range(4)]; Ssum = 0; bad = []; okrec = True
for n in range(1, NMAX + 1):
    k = n - 1
    if k >= 4:
        m = k - 4
        q, r = divmod(-(p0(m)*win[0] + p1(m)*win[1] + p2(m)*win[2] + p3(m)*win[3]), p4(m))
        okrec &= (r == 0); win = win[1:] + [q]; ak = q
        if k < 400: okrec &= (ak == A[k])
    else:
        ak = win[k]
    Ssum += (20*k + 17)*ak
    if Ssum % n: bad.append(n)
report("Sun's conjecture n | S(n) for all n<=%d (recurrence cross-checked); %.0fs" % (NMAX, time.time()-t0),
       okrec and not bad)


# 9b. second-layer reduction (Theorem 5.2): S(p^2 m) = S(p^2) * sum_{t<m} a(t) mod p^2,
#     and S(p^2) = -(3 w_{p^2-1} + 20 w_{p^2-2}) mod p^2
NL = 4000
AL = [a_direct(n) for n in range(4)]
for m in range(NL):
    q, r = divmod(-(p0(m)*AL[m] + p1(m)*AL[m+1] + p2(m)*AL[m+2] + p3(m)*AL[m+3]), p4(m)); AL.append(q)
SL = [0]; PL = [0]
for k in range(NL): SL.append(SL[-1] + (20*k + 17)*AL[k]); PL.append(PL[-1] + AL[k])
okL2 = all((SL[p*p*m] - SL[p*p]*PL[m]) % (p*p) == 0
           for p in sp.primerange(2, 64) for m in range(1, NL//(p*p) + 1) if p*p*m <= NL)
okL2b = all((SL[p*p] + 3*w(p*p-1) + 20*w(p*p-2)) % (p*p) == 0 for p in sp.primerange(2, 24))
report("Thm 5.2: second-layer reduction (p<64, p^2 m<=%d) and S(p^2) = -v_{p^2-1} mod p^2 (p<24)" % NL,
       okL2 and okL2b)

# 10. Dwork-type congruences (Conjecture 7.1): S(pn) = lam_p S(n) mod p^{v_p(n)+1}
NN = 1500
AA = [a_direct(n) for n in range(4)]
for m in range(NN):
    q, r = divmod(-(p0(m)*AA[m] + p1(m)*AA[m+1] + p2(m)*AA[m+2] + p3(m)*AA[m+3]), p4(m)); AA.append(q)
SS = [0]
for k in range(NN): SS.append(SS[-1] + (20*k + 17)*AA[k])
def vp(x, p):
    if x == 0: return 10**9
    e = 0
    while x % p == 0: x //= p; e += 1
    return e
okD = True
for p in sp.primerange(3, 32):
    base = 1 if p != 17 else 2
    mod = p**10
    lam = SS[p*base]*pow(SS[base], -1, mod) % mod
    okD &= vp(lam, p) >= 1
    okD &= all((SS[p*n] - lam*SS[n]) % p**(vp(n, p) + 1) == 0 for n in range(1, NN//p))
report("Conjecture 7.1 (Dwork-type) numerically (3<=p<=31, pn<=%d)" % NN, okD)

print("ALL CHECKS PASSED" if ok_all else "SOME CHECK FAILED")
```

References

[1] F. Beukers and M. Vlasenko, Dwork crystals I, *Int. Math. Res. Not. IMRN* 2021, 8807–8844.

[2] F. Beukers and M. Vlasenko, Dwork crystals III: from excellent Frobenius lifts towards supercongruences, *Int. Math. Res. Not. IMRN* 2023, no. 23, 20433–20483; arXiv:2105.14841.

[3] Google DeepMind et al., *formal-conjectures*, file `FormalConjectures/OEIS/179524.lean`, commit of 2026-11-06, https://github.com/google-deepmind/formal-conjectures.

[4] J. Guillera, Some binomial series obtained by the WZ-method, *Adv. in Appl. Math.* 29 (2002), 599–603.

[5] V. J. W. Guo and J. Zeng, Proof of some conjectures of Z.-W. Sun on congruences for Apéry polynomials, *J. Number Theory* 132 (2012), 1731–1740.
[6] V. J. W. Guo and W. Zudilin, A q-microscope for supercongruences, *Adv. Math.* 346 (2019), 329–358.
[7] D.-W. Hu and G.-S. Mao, On an extension of a Van Hamme supercongruence, *Ramanujan J.* 42 (2017), 713–723.
[8] Keio, Erdős's conjecture on $\sum f(n)/n$ is false: a 2-adic shadow of Okada's criterion and the least counterexample $q = 105^2$ (24 October 2026), item [055] of the Verifiable-Mathematical-Reasoning-Corpus, https://github.com/QEEDDD/Verifiable-Mathematical-Reasoning-Corpus.
[9] Keio, Square moduli, sparse masks, and sign obstructions for vanishing Erdős sums (research draft, 24 October 2026), item [056] of the same corpus.
[10] Keio, Erdős's conjecture on $\sum f(n)/n$ holds for every non-square modulus (24 October 2026), item [057] of the same corpus.
[11] A. Mellit and M. Vlasenko, Dwork's congruences for the constant terms of powers of a Laurent polynomial, *Int. J. Number Theory* 12 (2016), 313–321.
[12] OEIS Foundation, The On-Line Encyclopedia of Integer Sequences, sequence A179524 (comments by Z.-W. Sun, July 1, 2010), https://oeis.org/A179524.
[13] T. Adamczewski, OEIS Open: How many conjectures can language models turn into theorems?, arXiv:2608.11941 (2026).
[14] Z.-W. Sun, Products and sums divisible by central binomial coefficients, *Electron. J. Combin.* 20(1) (2013), #P9; arXiv:1004.4623.
[15] Z.-W. Sun, Open conjectures on congruences, *Nanjing Univ. J. Math. Biquarterly* 36 (2019), 1–99; earlier versions arXiv:0911.5665.
[16] W. Zudilin, Ramanujan-type supercongruences, *J. Number Theory* 129 (2009), 1848–1857.

Drafted with Claude (Anthropic) in a Claude Code session; all computations are reproducible with the script in Appendix A.